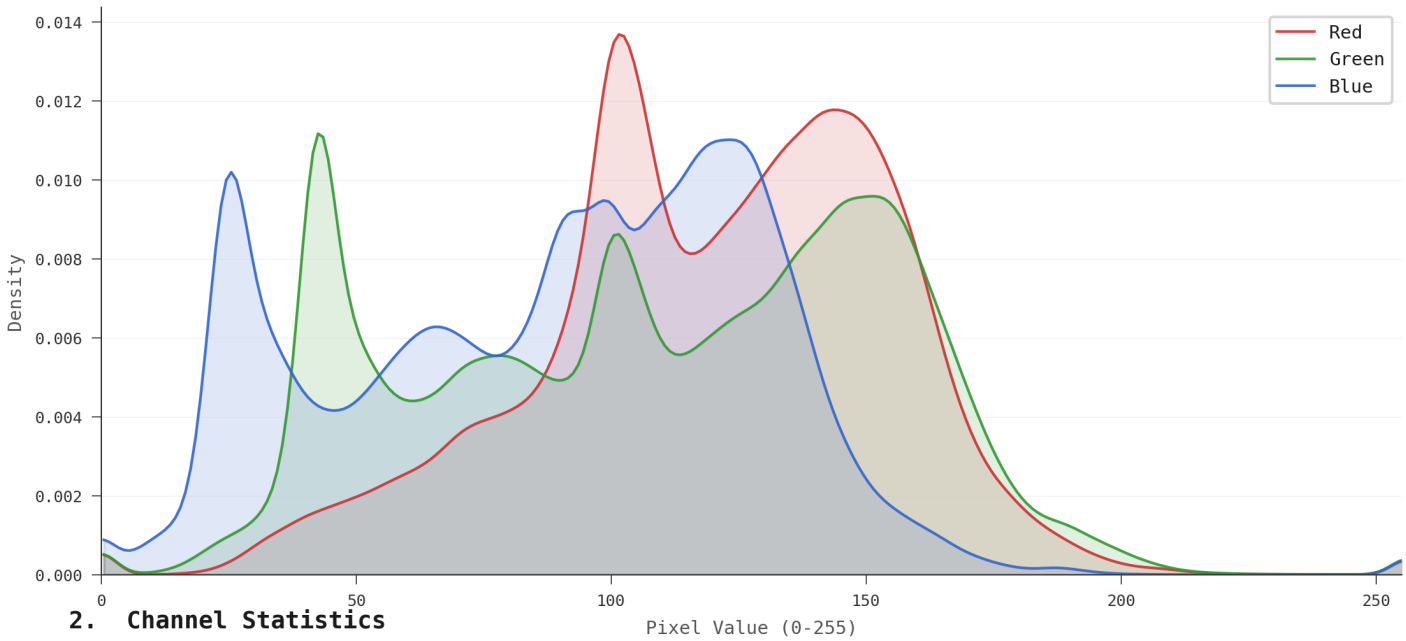
Reference Data Sheet: KODIM01

Subject: Brick Building with Red Shutters and Door

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	119.7613	34.5326	1192.5027	0.1725	-0.3106	0	255
Green	108.7463	44.7298	2000.7507	-0.9301	-0.1048	0	255
Blue	88.3865	40.1137	1609.1105	-0.5851	-0.1370	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8627	0.8400
G	0.8627	1.0000	0.9914
B	0.8400	0.9914	1.0000

Pairwise Pearson Correlation Coefficients

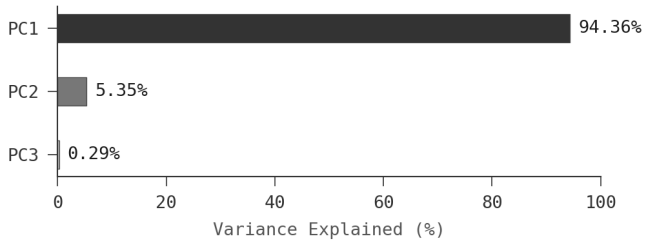
r(R-G)	=	**0.862705**
r(R-B)	=	**0.840004**
r(G-B)	=	**0.991359**
Avg |r|	=	**0.898023**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1192.5057} & 1332.5684 & 1163.6043 \\
1332.5684 & \mathbf{2000.7558} & 1778.7776 \\
1163.6043 & 1778.7776 & \mathbf{1609.1146}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM01

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.4681	**-0.6597**	-0.5879	4531.6251
PC2	-0.8806	0.2927	0.3727	257.0119
PC3	0.0738	-0.6922	0.7179	13.7391

6. Derived Metrics

Condition Number

329.83

(l1/l3) PC1/PC2: 17.63 | PC2/PC3: 18.71

Blue Channel Independence

2.7%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6597 | Green-Blue coupled

Classification: Strongly One-Dimensional (PC1 = 94.36%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.855063	0.785762	0.820413
Green	0.904122	0.864049	0.884085
Blue	0.886151	0.843665	0.864908

8. Average Local Variance (3x3)

Channel	Value
Red	326.7876
Green	367.5441
Blue	335.7166

9. Redundancy Profile

Classification: Strongly One-Dimensional | Condition Number: 329.83 | PC1: 94.36%

Correlation: G–B pair dominates at r = 0.991359 | R–G: 0.862705 | R–B: 0.840004

Blue Channel: 2.7% independent | 97.3% captured by PC1

Spatial: Avg autocorrelation: 0.856469 | Avg local variance: 343.3494

Eigenvector: Green-Blue coupled | G (-0.6597), B (-0.5879) co-load | R offset (-0.4681)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."